

Thais Isabelle Parron Ruiz

Tampa, FL | +1 (656) 215-5187 | thaisisabelle.parron@gmail.com | [/in/thais-parron-ruiz](https://github.com/thais-parron-ruiz) | thaisparron.github.io/portfolio

EDUCATION

University of South Florida - Tampa, FL
Bachelor of Science in Mechanical Engineering
Minor: Aerospace Engineering

GPA: 3.87
Expected Graduation Date: Dec 2026

EXPERIENCE

Engineering Tutor

AUG 2024 – Present

University of South Florida

Tampa, FL

- Tutor at the Engineering Student Success Center in Statics, Dynamics, Thermodynamics, Electrical Systems I, programming, Computational Methods, and Kinematics & Dynamics of Machinery.
- Assist an average of 2 students per hour across all supported subjects.

Water Harvesting Research

AUG 2024 – DEC 2025

University of South Florida

Tampa, FL

- Designed a sustainable water collection device using laser-modified aluminum with alternating superhydrophilic and superhydrophobic surface patterns to enhance droplet formation and flow.
- Integrated a top-mounted fan for airflow and a central cooling system to accelerate condensation, improving collection efficiency in humid, low-resource environments.

Volunteer

JAN 2020 – DEC 2021

Rotary – Interact Club

Cocoa, FL - BRAZIL

- Helped with planning and organizing 5 fundraising events where we collected \$10k in a year.
- Encouraged donations by supporting community fundraising events and campaigns.

ENGINEERING PROJECTS

Dual-Action Can Crusher Design

JAN 2025 – MAY 2025

Kinematics & Dynamics of Machinery Final Project

- Designed a dual-action can crusher using a 6-bar linkage in SolidWorks. Reduced input force by 40% through horizontal pre-denting before vertical crushing.
- Built a refined functional prototype using aluminum and wood, resolving interference with spacers and recommending steel for improved stiffness.

Projectile Simulation Game

MAR 2024 – APR 2024

Programming Concepts

- Developed a projectile-launching game in MATLAB, where players aimed to hit a target at the base of a mountain with a precision of within 2 meters, incorporating physics-based calculations for trajectory and launch dynamics.
- Implemented dynamic visuals in the game, displaying the mountain, target, and projectile trajectory, while allowing players multiple attempts.

SOLIDWORKS Engineering Project

AUG 2023 – DEC 2023

Computer Aided Design and Engineering

- Designed an acoustic guitar in SolidWorks, utilizing multiple features to model its components accurately.
- Created functional tuning pins that mimic real-life mechanics, which would allow string tension adjustment.

Action Figure Bot

JAN 2023 – MAY 2023

Foundations of Engineering Laboratory

- Designed in a group of 5 students, an object follower robot with 3D printed chassis to introduce 100+ K-12 students to the Engineering and Robotics field.
- Built a prototype that utilized 1 Ultrasonic sensor to follow an object by coding in Arduino.

Sustainable Engineering Project

JAN 2023 – MAY 2023

Engineering Economics with Social and Global Impact

- Created, in a group of five students, a Sustainable Engineering project for Hillsborough County.
- The project aimed to protect the Hillsborough River through dams, reservoirs, and aquifer replenishment, focusing on community well-being and wildlife preservation.

SKILLS

Software: SOLIDWORKS, MATLAB, Simulink, LabVIEW, Fusion, LogixPro PLC Simulator, Microsoft Office.

Languages: English (fluent), Portuguese (native), Spanish (basic).

Certifications: Mechanical Design and Additive Manufacturing from SOLIDWORKS.